THE LIMITATIONS OF EQUILIBRIUM CONCEPTS IN EVOLUTIONARY GAMES

AYDIN MOHSENI

Abstract. Within the framework of evolutionary game theory, equilibrium concepts adapted from rational choice game theory are employed to identify the probable outcomes of evolutionary processes. Over the past several decades results have emerged in the literature demonstrating limitations to each of the proposed equilibrium concepts. These results rely on an undefined notion of evolutionary significance. We explicitly define evolutionary significance for evolutionary games. This definition enables an analysis of the success of equilibrium concepts across different models of evolution. We demonstrate that even under favorable assumptions as to the underlying dynamics and stability concept—the replicator dynamics and asymptotic stability—each equilibrium concept makes errors of either omission or comission; often both.

1. Introduction

The search for an equilibrium concept in evolutionary games theory (EGT) has been driven by the hopes for an effective, general, and theoretically simple method of analysis. Forty years since the introduction of evolutionarily stable strategy (ESS) concept by Maynard Smith and Price (1973), it seems that no such equilibrium concept is to be had, and this absence has become a part of the folk wisdom of the community of practitioners of EGT. Yet, in certain areas of ecology and evolutionary biology, it is not uncommon to find the exclusive use of the equilibrium concepts, the leading candidate among which is the still the ESS.

A search for "Ecology" and "ESS" in Google Scholar reveals that 13 of the 20 top papers since 2015 that used the method of ESS analysis used it exclusively. That is, they followed what Huttegger and Zollman have termed "ESS methodology" (2010). These papers describe an evolutionary phenomenon in terms of a game, find the ESS or ESSs of that game, and equate these with the possible outcomes of evolution. This can be problematic. As we will discuss, there are a breadth of conditions under which each of the primary equilibrium concepts make serious errors of both omission and commission. Indeed, there are in-principle reasons why we can expect equilibrium concepts to come up short. This paper presents a story of how and why this is so, and, in doing so, presents a novel, explicit characterization of evolutionary significance for dynamical models of evolution.

Date: June 5, 2018.

In §2, we introduce the concepts of evolutionary games and dynamics needed to understand our methodology and results. In §3, we unpack the notion of "evolutionary significance" until we reach a formulation that lends itself to being made mathematically precise and guide our evaluations of the success and failure of equilibrium concepts. In §4, we present the stability concepts that, with respect to a given family of dynamics, can formalize our intuitions about evolutionary significance. In §5, we introduce a precise characterization of equilibrium concepts, and examine their relationship to stability concepts.

In §6, we present the assumptions and methodology underlying our analysis. We defend the use of the replicator dynamics (RD) and asymptotic stability (AS) as our models of evolution, and condition for evolutionarily significance, respectively. We show that the RD and AS constitute assumptions favorable to the success of the equilibrium concepts, and so their failure with respect to these assumptions are suggestive of more general limits of the concepts.

In §7, we circumscribe the limitations of the main equilibrium concepts used to analyze evolutionary games. We consider the Nash equilibrium (NE), strict Nash equilibrium (strict NE), evolutionarily stable strategy (ESS), neutrally stable strategy (NSS), and evolutionarily stable set (ES set) concepts. We show how each equilibrium concept predicts outcomes that we would not consider evolutionarily significant, and each fails to predict what are clearly evolutionarily significant outcomes. We also provide a distinct in-principle argument for expecting that any point- or set-valued equilibrium concept will fail in capturing all significant outcomes.

In §8, we discuss these results with an eye to their implications for the challenges faced by any viable method of analysis for evolutionary games. In §9, we conclude.

2. Evolutionary Games and Dynamics

An evolutionary game can be thought of as composed of two parts: a game, and an underlying evolutionary dynamics. The game describes the interaction structure of a population in conflict, cooperation, signaling, mating, and so on. The dynamics describes our hypothesis as to the behavior of evolution: things like the nature of selection, mutation, and drift will be specified in the dynamics; population size and structure will be accounted for; whether interactions between the different individuals in the population are random or correlated; whether reproduction is sexual or asexual, that is, the details of transmission will be specified here; and even exogenous factors, such as environmental variability will typically be accounted for in the dynamics.

Evolutionary games are used in a breadth of contexts—i.a., animal behavior, ecology, epidemiology, oncology, economics, and anthropology—where players can be interpreted variously as representing genes, individuals (bacteria, people, and so on), or memes. For simplicity, we stick to largely biological interpretations and vocabulary where players

	s_1	s_2
s_1	a_{11}	a_{12}
s_2	a_{21}	a_{22}

FIGURE 1. Payoff matrix for a simple 2×2 symmetric normal form game.

are taken to represent individual organisms, strategies are to be understood as behavior patterns or phenotypes, and payoffs as contributions to Darwinian fitness or reproductive success.

We can describe a game using a payoff matrix A specifying the possible outcomes of interactions between combinations of strategies in terms of the resulting contributions to the fitness of each strategy involved. For example, consider the two-player symmetric game given in FIGURE 1. If one member of the population uses strategy s_1 against an individual using strategy s_2, the matrix specifies that the resulting contributions to their fitnesses are $A(s_1, s_2) = a_{12}$ and $A(s_2, s_1) = a_{21}$, respectively.

A mixed strategy is a weighted combination of strategies with the weights summing to one. Looking at the two strategy game presented in TABLE 1, a mixed strategy is any strategy $s = x \cdot s_1 + (1 - x) \cdot s_2$ with $x \in [0, 1]$. In the context of an evolutionary game, s may be interpreted as reflecting a monomorphic (single type) population where individuals perform different actions with probabilities x and $(1 - x)$, or as reflecting a polymorphic (multi-type) population where the mixture reflects the proportion of the population playing each of the available pure strategies, s_1 and s_2. For simplicity we stick with the latter interpretation throughout, and take mixed strategies to represent population proportions of types.[1]

More generally, a symmetric normal form game G is specified by a quadruple $\langle P, S, \Delta^n, A \rangle$ composed of: a set of players involved in the interaction $P = \{1, \ldots, m\}$; the set of pure strategies $S = \{s_1, \ldots, s_n\}$ corresponding to possible phenotypes in the population; the state space $\Delta^n = \{\langle x_1, \ldots, x_n \rangle \mid x_i \geq 0, \sum x_i = 1\}$ of all possible states of a population, and a payoff matrix $A = [a_{ij}]$ specifying the resulting payoffs to each player from each combination of strategies. The expected payoff to a population mixture i composing proportion x_i of the population in interaction with other population mixtures (say, of invading mutants) is given by $u(x_i, x_{-i})$.

For the results considered here, we restrict our interest to a relevant class of games. Namely, we consider finite games, where interactions are between a finite numbers of players who have a finite number of possible (pure) strategies available to them. We

[1]For discussion of the various interpretations of mixed strategies and their implications see Bergstrom and Godfrey-Smith (1998) and Osborne (2000, 37-42).

consider both simultaneous and sequential games. For the example in this paper, we will typically use symmetric normal form two- and three-strategy games.

The second component of an evolutionary game is the evolutionary dynamics. The dynamics translate the resulting payoffs of these interactions between strategies into changes in the proportions of the population using the different strategies. The leading idea is that, with the introduction of dynamics, the proportions of the strategies in the population affect the payoffs to each strategy, and so, fitness becomes frequency dependent.

For our analysis, we employ the "first and most important model of evolutionary game theory" (Cressman and Tao 2014, 1081), the replicator dynamics (RD). The leading idea behind RD is that the growth rate of each strategy is a function of the difference of the strategy's fitness and the population's mean fitness. A strategy whose fitness is greater than the population average grows in relative proportion, and one whose fitness is less than average shrinks. Formally, the continuous time RD[2,3] consists of a system of differential equations,

$$\dot{x}_i = x_i \left[u(i, x) - u(x, x) \right],$$

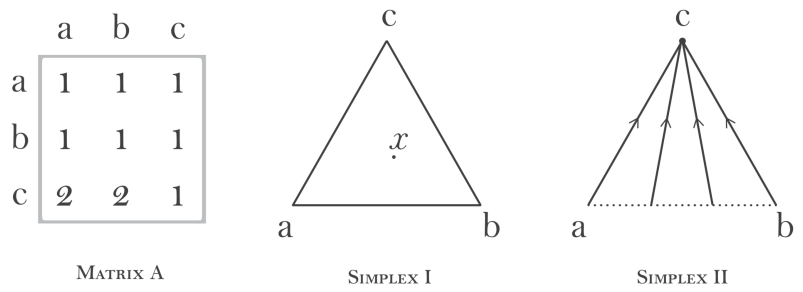
where $\dot{x}$ denotes the rate of change of the population proportion of strategy i, x_i denotes the proportion of strategy i, $u(i, x)$ denotes the expected fitness of type i from interaction with the population, $u(x, x)$ denotes the population mean fitness, and x denotes the composition of the population as a whole.

An equilibrium, then, is just a fixed or stationary point of the dynamics. In the continuous time replicator dynamics, a state x is a *fixed point* of the dynamics if $\dot{x}_i = 0$ for all i. Not all equilibria are made equal. Some sets of equilibria will, either individually or collectively, constitute plausible outcomes of a dynamics, while others will not. As we will see, the measure of a successful equilibrium concept is in making just such a distinction, and selecting the right subset of possible outcomes. For reasons we will discuss, the challenge here is daunting.

3. Capturing Evolutionary Significance

To appreciate the results discussed in this paper, we need a precise understanding of how to evaluate the success of equilibrium concepts in the context of evolutionary games. This is typically explained in terms of 'capturing *evolutionary significance*'. That is, it is assumed that for an equilibrium concept to be successful it must capture all and only the states of an evolutionary process that are probable outcomes of that process.

[2]For discussion of discrete and continuous time formulations of RD see (Schuster and Sigmund 1983).
[3]The results we consider here employ the continuous time RD exclusively.



Matrix A Simplex I Simplex II

FIGURE 2. Payoff matrix and phase portrait of a dynamics with a single probable outcome.

The notion of evolutionary significance has been used a number of times in the EGT literature (Skyrms 2000; Huttegger and Zollman 2010, 2013), but has yet to be explicitly defined. Our challenge is that evolutionary significance has been, to-date, an essentially informal desideratum. We attempt to provide such a definition in the context of evolutionary games, and in doing so, arrive at a novel probabilistic account of evolutionary significance. Our definition remains true to the established framework of EGT, while sketching clearer connections between the intuitive notion of evolutionary significance, and the formal definitions of equilibrium concepts stemming from game theory and stability concepts stemming from dynamical systems.

To begin, what we want to say is something to the following effect: that the significant outcomes of an evolutionary dynamic are just those that are the probable outcomes of that dynamic. At first pass, this yields the characterization:

Characterization 1 (Evolutionarily Significant Outcome)**.** *Given an evolutionary game and its dynamics, an outcome is evolutionarily significant if it is a probable outcome of that dynamics.*

But this definition needs be unpacked and sharpened in turn, and we must continue unpacking and sharpening the assumptions implicit in evolutionary significance until we arrive at those amenable to precise formulation. The result of this process will be to provide us with the criteria needed to evaluate the success of equilibrium concepts. First, we may unpack the notion of *probable outcome*.

To illustrate the concepts involved, we can proceed by use of the example of an evolutionary game which exhibits a single probable outcome. Let the simplex in FIGURE 2 be the state space of a simple 3-strategy symmetric evolutionary game. Each point in the simplex represents a possible population state—a vector $x = \langle x_a, x_b, x_c \rangle$ specifying the population proportions of the strategies a, b, and c. The three vertices represent the monomorphic states where the population is composed entirely of a single strategy, so the a-vertex corresponds to the state vector $\langle 1, 0, 0 \rangle$, and so on. Distance from a vertex corresponds inversely to the population proportion of the strategy represented by the vertex.

Given a game, the dynamics specify how the state vector changes over time, moving it through the state space of possible population proportions. The path that a population moves along in the state space as it evolves is called its trajectory. Which trajectory a population takes depends on its initial conditions. We visualize the "flow" induced on the state space by the replicator dynamics by drawing representative trajectories where the directions of flow are designated by using arrows as in FIGURE 2. Diagrams such as Simplex II in FIGURE 2 may be used to represent our game dynamics and are called phase portraits. Here, dotted lines represent sets of fixed points, opaque circles represent sinks, and empty circles represent sources.

In this game, we would clearly want to say that the state in which strategy c dominates the population is the sole probable outcome of evolution. Our reasoning is as follows. We begin by assuming that it is there is some positive probability that population will start in any open interval of the state space.[4] (For example, in the simplest case, this can be a uniform distribution.) It follows from this stipulation that we must assign a continuous probability distribution over every point in the state space with respect to the initial conditions of the dynamic.

Next, we observe that the dynamics indicate that from *most any* starting point, the population will evolve toward the all-c vertex of our simplex, where the population is composed entirely of individuals using strategy c. We further observe that, from any state where the proportion of strategy c is nonzero, the dynamics will converge, in the limit of time, to the state where the population proportion of strategy c is one. Thus, it appears that the state $x^* = \langle 0, 0, 1\rangle$ where only strategy c is present is a probable outcome of the game.

We can further develop our concept of a probable outcome by considering what may constitute an *improbable outcome*. Consider the dotted edge $\overline{ab}$ at the bottom of the simplex: this line corresponds to the population mixtures of a and b where strategy c is wholly absent. At all population states along this line, excepting its endpoints, strategies a and b receive the same fitness. There is no selective pressure in either direction since

$$u_a = x \cdot A(a,a) + (1-x) \cdot A(a,b) = 1 = x \cdot A(b,a) + (1-x) \cdot A(b,b) = u_b,$$

where x and $(1-x)$ denote the proportions of strategies a and b, respectively. Given that the fitness of both strategies is equal, it seems that a population starting at this edge would have no reason to leave, and so would remain here for all time.

Given a continuous distribution over initial conditions, however, the chances of actually starting on this edge are negligible, since any one-dimensional edge composes a

[4]To do so, we assume the Lebesgue measure on the n-dimensional volume of the state space $\Delta^n = \{\langle x_1, \ldots, x_n\rangle \,|\, x_i \geq 0, \sum x_i = 1\}$, where $n \equiv |S|$ denotes the cardinality of the strategy set of the game.

measure-zero subset of our two-dimensional space.[5] By this reasoning, the set of states composing $\overline{ab}$ constitute improbable initial conditions of the population, and so, are not a probable outcome of our dynamic.

But what if our priors are not continuously distributed on the whole state space? Imagine that we know that the population will start at one of the population states represented by the edge $\overline{ab}$. In this case, we may ask again: does any portion of the edge $\overline{ab}$ compose a probable outcome of our evolutionary dynamics? Perhaps surprisingly, our conclusion must remain that no subset of the edge is a probable outcome of evolution. The reason for this is a "hidden mutation assumption." That is, while it is not explicitly represented in the replicator dynamics, we assume that there is some possibility of invasion by mutant. Small numbers of any alternative strategy may be produced by mutation and lead to small but persistent fluctuations in the population proportions. Here, any small perturbations of the population state from the edge would cause the population to head off toward the top of the state space, never to return. The inevitable presence of mutant strategies and natural fluctuations means that an outcome cannot be a probable one unless it is stable under small perturbations.

We now have a definition of probable outcome in terms of convergence from a significant set of initial conditions, and resistance to small perturbations.

Characterization 2 (Probable Outcome)**.** *An outcome is probable if, given a continuous probability distribution on initial conditions of the state space, a set of initial conditions of positive measure converge to the outcome, and, upon convergence, are stable in the face of small perturbations.*

But we must also be precise about what constitutes an *outcome*? In order to provide a principled and unifying formulation of what constitutes an outcome of a nonterminal process, we must lean on the notion of limits. An outcome, then, becomes a possible behavior of the system in the limit of time. Thus we have

Characterization 3 (Outcome)**.** *An outcome is a possible behavior of an evolutionary game in the long run, as time goes to infinity.*

But there is still something more we want to say. Outcomes should not be limited to convex sets of states. Outcomes should include (sets of) closed orbits. For example, in the rock paper scissors (RPS) game in FIGURE 3, the limit behavior of the dynamics is a continuum of cycles around the center point of the state space, and any point on the interior of the simplex is a part of some cycle for some initial conditions of the system. So while no single orbit is a probable outcome, the continuum of orbits is. What we have is that sets of closed orbits, and not only sets of states, may be outcomes of a dynamics.

[5]More generally, any $(n-1)$-dimensional subspace of an n-dimensional state space will be assigned measure zero.

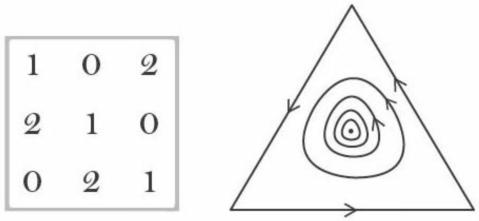


FIGURE 3. Rock-paper-scissors game exhibiting cycles.

This allows us to capture key disequilibrium behavior such as limit cycles and strange attractors that may constitute persistent long run outcomes of dynamics (Skyrms 1992b; **?**). With this, we have arrived at formal conditions by which to evaluate equilibrium concepts. We combine characterizations 2, and 3 to get:

Characterization 4 (Evolutionary Significance)**.** *A set of states or closed orbits is a evolutionarily significant if, given a continuous probability distribution on initial conditions of the state space, a set of initial conditions of positive measure converge to the set in the limit of time, where they are stable in the face of small perturbations.*[6]

We have characterized evolutionary significance in terms of probable outcomes under some evolutionary dynamics, and probable outcomes as states (or closed orbits) that induce convergence from a significant set of initial conditions, and exhibit stability under perturbations. This is what we want, as we can mathematically formalize the concepts of *attraction* and *stability* in the form of our stability concept, and we can then use the stability concept to evaluate the success and failure of equilibrium concepts. For our results, this will end up being asymptotic stability.[7]

Having arrived at a sharper definition of evolutionary significance for this context, we can return to the simple evolutionary game in FIGURE 2 to conclude that the single probable outcome of this evolutionary game is the monomorphic population state $x^* = \langle 0, 0, 1\rangle$ where strategy c has come to fixation. The state x^* is evolutionarily significant because it commands convergence from a probabilistically significant set of initial conditions composed of the interior of the simplex, and any population already at x^* is stable. There are no other evolutionarily significant (sets of) states, as any points not on in the interior of the simplex is on its edges, and the edges of the simplex compose a measure-zero subset of the state space, and so do no constitute probable initial conditions.

[6]We should note that this allows for evolutionary significance to be relativized to known initial conditions that may compose strict subsets of the state space. In practice, if we have an informed probability distribution over initial conditions (e.g., inferred from the biological record or determined by experimental controls), we can speak of probable outcomes relative to these initial conditions. It can be shown that restricting the set of initial conditions with positive probability can eliminate, but not add, to the set of evolutionarily significant outcomes of a game.

[7]Though as we will see, this will not quite suffice to capture (sets of) closed orbits.

4. Stability Concepts

We have provided a sharper definition of evolutionary significance so that we can have clarity about the connection between our intuitive notion of evolutionary significance and mathematical precise formulations of this notion. These formulations are *stability concepts*.

First, we note that a basic fact about stability concepts is that they are dynamics-specific. Stability concepts are defined in relation to dynamics, or classes of dynamics. For our results, we will define the stability concepts for the replicator dynamics. The main candidate stability concepts are *asymptotic stability*, and *Lyapunov stability*.

Definition 1 (Lyapunov Stable)**.** *A fixed point $x^* \in \Delta$ is stable if points near it remain near it.*

Definition 2 (Attracting)**.** *A fixed point $x^* \in \Delta$ is attracting if nearby points tend toward it.*

Definition 3 (Asymptotically Stable)**.** *A fixed point $x^* \in \Delta$ is asymptotically stable if it is both stable, and attracting.*[8]

The stability concept provides criteria for evolutionary significance in relation to the game dynamics. By being defined in terms of the dynamics, the stability concept provides a richer, more reliable but less general notion than that which can be provided by an equilibrium concept.[9] It becomes clear then, that if the underlying dynamics is taken seriously, the goal of the equilibrium concept should be to pick out essentially the same (sets of) states as the stability concept.

5. Equilibrium Concepts

Equilibrium concepts were imported into evolutionary game theory from classical, rational choice game theory. An essential insight that has motivated the adaptation of equilibrium concepts into evolutionary game theory is that evolutionary processes can be thought of as solving optimization problems in a way not entirely dissimilar to the recommendations made to rational agents in classical game theory. Where strategies maximizing individual expected utilities may deliver a group of rational agents to an equilibrium, evolutionary dynamics driven by differentials in reproductive or imitative success may deliver populations of non-rational, or bounded rationality, agents to stable states.

The candidate equilibrium concepts are: Nash equilibrium (NE), strict Nash equilibrium, evolutionary stable strategies (ESS), neutrally stable strategies (NSS), and

[8]For precise definitions of the stability concepts see (Cressman [2003], pp.18-19).

[9]Note that these stability concepts also omit some of the outcomes we want to deem as evolutionarily significant, such as cycles and strange attractors. However, for now, asymptotic stability gives us a sufficient and precise characterization of significant outcomes. We will return to exceptional cases in later sections.

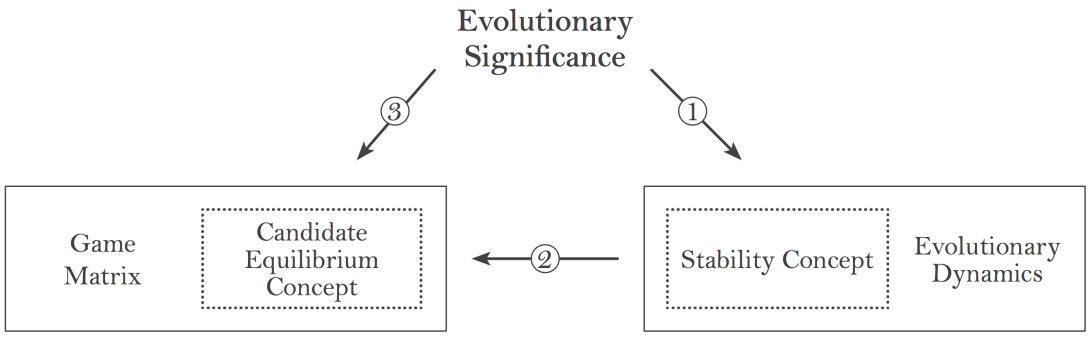


FIGURE 4. A diagram of the interrelationship of evolutionary significance, equilibrium concepts, and stability concepts.

evolutionarily stable sets (ESSets).[10] Each equilibrium concept is a refinement of the Nash equilibrium. That is, each starts with the set of Nash equilibria of a game and trims down the set by eliminating those equilibria it deems implausible (Huttegger et al. 2014). The exception to this is the ESSet concept. Rather than assessing individual equilibrium states, the ESSet picks out sets of states that, when considered collectively, exhibit equilibrium properties.

Formally, an equilibrium concept is a mapping $F : \Gamma \to \cup_{G\in\Gamma} 2^{\Delta_G}$ from the set of all games Γ to the union of the states spaces of all possible strategy profiles. That is, F takes as arguments a game G, and the set of the possible states Δ_G for that game (its state space, as previously defined), and outputs a subset $F(G, \Delta_G)$ of those states. For our applications to evolutionary games, the aim is that one of these subsets turns out to be just the evolutionarily significant outcomes of that game.

What is crucial to observe is that, whereas stability concepts are defined in relation to the evolutionary dynamics, equilibrium concepts are defined exclusively in relation to the interaction structure of the strategies. That is, equilibrium concepts are defined exclusively in terms of the game. That equilibrium concepts can be used to attempt to infer the plausible outcomes of evolution by considering only the game matrix—and without deferring to any specific underlying model of the evolutionary dynamics—is essential to both their theoretical simplicity and generality, and, ultimately, to their limitations and unreliability.

In FIGURE 4, the interrelationship of evolutionary significance, equilibrium concepts, and stability concepts is visualized. In (1), our informal notion of evolutionary significance must be made mathematically precise in the form of a stability concept, which must be defined with respect to some (class of) evolutionary dynamics. In (2), we can now evaluate the equilibrium concept, defined at the level of the game matrix, in terms

[10]One may consider other equilibrium concepts from game theory, such as the subgame perfecti equilibrium, trembling hand equilibrium, sequential equilibrium, and so on. However, for various reasons, they are not typically seen as candidates for evolutionary games, and so are not considered here.

of its agreement with the stability concept. This gives us, in (3), a way to indirectly assess the success of a given equilibrium concept in capturing evolutionary significance under a given class of matrices and dynamics. For us, these are finite games under the replicator dynamics.

In this way, equilibrium concepts explain evolutionary significance in so far as they are in agreement with the stability concept; we assess each equilibrium concept by its success in capturing the set of states that are asymptotically stable.

We have presented a novel, and more precise formulation of evolutionary significance that we argue captures what implicitly underpins determination of the success of equilibrium concepts. We have formalized this definition as a stability concept—asymptotic stability— that is defined in relation to a hypothesis as to the behavior of evolution—the replicator dynamics. With this, we have established a framework in which we can assess the viability of an equilibrium concept—in terms of its agreement with the stability concept, and hence with our conditions for evolutionary significance.

6. Methodology of Analysis

The results gathered in this paper collectively demonstrate that even under favorable assumptions regarding the underlying dynamics and standards of stability—replicator dynamics and asymptotic stability—all candidate equilibrium concepts fall short. The approach taken here, and in much of the literature, is to test the equilibrium concepts under assumptions constituting a "best-case scenario." The idea is that, if the equilibrium solution concepts fail under such assumptions, a sort of informal "upper bound" for the domain of their reliable application has been established. For these results to carry weight, we must justify why replicator dynamics and asymptotic stability constitute such a best case for our equilibrium solution concepts.

Among evolutionary dynamics, the replicator dynamics are commonly regarded as the simplest member of the class of deterministic dynamics (Sandholm 2010). Our claim is that if an equilibrium concept fails to capture evolutionary significance for the replicator dynamics, then it has little hope to do so for other more complicated dynamics. The reason for this is that RD make largely the same assumptions about the population as the ESS concept does (Apaloo et al. 2009). This should not come as a surprise, as the RD were formulated by Taylor and Jonker (Taylor and Jonker 1978) specifically for the purpose of providing an explicit evolutionary dynamic for the ESS concept. And as all of our candidate equilibrium concepts are refinements of the Nash equilibrium concept, just as ESS is, they carry largely the same assumptions as to the nature of evolutionary populations involved. Specifically, each of the equilibrium concepts can be thought of as implicitly assuming infinite populations, no drift, no population structure, and a uniform environment.

To see how these assumptions make the replicator dynamics congenial to our equilibrium concepts, consider two somewhat more complex dynamics—the replicator-mutator dynamics, and the Moran process. The replicator-mutator dynamics differ from RD only in that they make mutation explicit (Bomze and Burger 1995), while the Moran process provides a stochastic analogue to RD, allowing for the modeling of finite populations and the concomitant effects of drift (Moran 1962). In the former dynamics, sufficiently high mutation rates can obviate any equilibrium prediction by changing the position of equilibria, and potentially collapsing fixed points of the dynamics (Hofbauer and Huttegger 2007). In the latter, small population sizes can select even for strictly dominated strategies (Fudenberg and Imhof 2004), defying a basic assumption of the equilibrium concepts.[11] In both cases, for the same game, parameters not represented at the level of the game matrix—mutation rate and population size—can be changed so as to lead to qualitatively distinct outcomes. Equilibrium concepts take as input only the payoff matrix and state space of game, and so cannot be sensitive to parameters not reflected in these two objects. In contrast to the replicator-mutator dynamics and Moran process, the replicator dynamics, given any game, will output a qualitatively unique evolutionary process (Weibull 1997). The RD are, by design, highly congenial to our equilibrium concepts.

Similarly, asymptotic stability is the most common stability concept used in tandem with deterministic dynamics, generally, and the replicator dynamics, specifically (Hofbauer and Sigmund 1998; Cressman 2003; Sandholm 2010). Stronger stability conditions—those that pick out fewer (sets of) states as stable—will cause the equilibrium concepts to perform worse, as the equilibrium concepts are already too weak with respect to asymptotic stability, and pick out many states it deems as unstable. Similarly, under the weaker assumption of Lyapunov stability it is known that the RD's performance is even worse (Bomze 1983, 1995). These results suggest that asymptotic stability may provide something in the neighborhood of locally optimal stability conditions for the replicator dynamics.

7. The Limitations of Equilibrium Concepts

We can now present an overview of the existing results on the limitations of equilibrium concepts in light of a unifying definition of evolutionary significance. We provide citations of original results where appropriate. A natural place to begin our overview and analysis is at the Nash equilibrium (NE) concept. This is because, in their logical structure, all of the equilibrium concepts we will be considering here are refinements of the NE concept. That is, each equilibrium concept can be thought of as beginning with the full set of Nash equilibria, and proceeding to eliminate those equilibria it deems are

[11] That is, a strictly dominated strategy can never be a best response, and so cannot be a part of a Nash equilibrium.

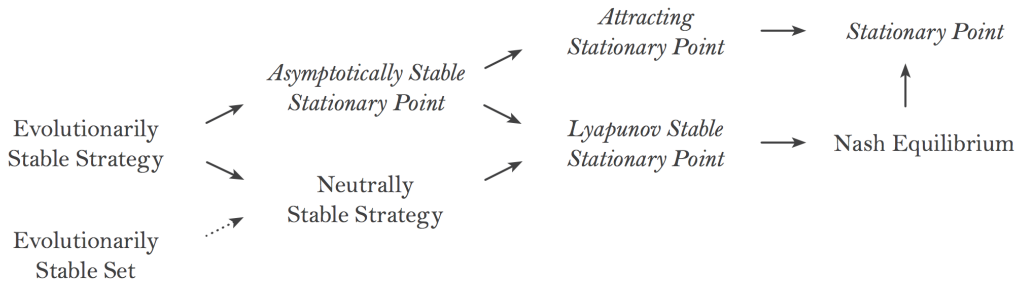


FIGURE 5. The flow of the implication relations between equilibrium concepts and stability concepts.[12]

unviable. The exception to this narrowing is the ESSet concept as it also expands the range of equilibrium states to allow for sets of states that are collectively (though not necessarily individually) asymptotically stable.

Each of these Nash refinements—the strict NE, ESS, NSS, and ESSet—will be shown to be, in one or more ways, too weak, in that they pick out (sets of) states that are not evolutionarily significant. Some equilibrium concepts will also be shown to be too strong in that they fail to pick out evolutionarily significant (sets of) states. Additionally, we will find that all equilibrium concepts share a way in which they are uniformly too strong: they all fail to capture intuitively significant outcomes, such as limit cycles and strange attractors, that are not expressible in terms of sets of states.

7.1. **Limits of the Nash Equilibrium and Strict Nash Equilibrium.** A conceptual framing of the Nash refinements project can be given by the dual facts that the Nash equilibrium is clearly far too weak, and that the strict Nash equilibrium is clearly far too strong. We will see that where the former includes in its predictions states which are neither stable nor attracting, the latter omits states which are both. In accounting for these facts, the thinking has been that an equilibrium concept intermediate in strength may pick out all and only evolutionary significant outcomes. For several reason we will examine—including the existence of stable orbits, and collectively but not individually stable states—this turns out to be infeasible.

The Nash equilibrium is the fundamental equilibrium concept of classical, rational choice game theory. A state is a NE in which no individual stands to gain by unilaterally switching her strategy—i.e., by switching her strategy when all the strategies of all the other players remain fixed. For a population in such a state, no individual has positive incentive to deviate, and so it will be a fixed point. In other words, if there are no perturbations, a population starting at that state will remain at that state. Formally, we define the NE as follows:

Definition 4 (Nash Equilibrium)**.** *A state $x^* \in \Delta$ is a Nash equilibrium if, for all i such that $x_i \in \Delta$, $u(x_i^*, x_{-i}^*) \geq u(x_i, x_{-i}^*)$.*

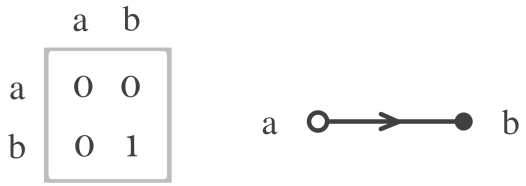


FIGURE 6. The all-a state $\langle 1, 0\rangle$ is a Nash equilibrium of the game, but is unstable under the replicator dynamics.

where $u(x_i, x^*_{-i})$ is the payoff to an individual using the (mixed) strategy x_i given that all others players $-i$ are using their Nash equilibrium strategies x_{-i}. For notationally clarity, we formulate our definitions from here on for the case of symmetric pairwise interactions (extension to games with more players should be clear). The NE then states:

Definition 5 (Nash Equilibrium)**.** *A state $x \in X$ is a Nash equilibrium if, for all $y \in \Delta$ where $y \neq x$, $u(x, x) \geq u(y, x)$.*

What is important to note about the Nash equilibrium solution concept is that all NE are fixed points under RD. The converse, however, is not true. Not all fixed points are NE. This is because, under RD, every unused strategy remains unused, and every pure strategy corresponds to a state of the population in which all but one strategy is unused. Hence, that single strategy will be the only strategy present for all future times as its population proportion will remain constantly at one. Thus, every pure strategy is a fixed point of the dynamics. However, all interior fixed points are indeed NE (Weibull 1997).

To demonstrate that the Nash equilibrium provides conditions too weak for evolutionary significance, we need only observe how fixed points that are neither stable nor attracting (these are known as sources) may be NE. Consider the simple symmetric 2×2 game in FIGURE 6. Both the monomorphic populations—those composed entirely of strategies a or b, corresponding to the state vectors $\langle 1, 0\rangle$ and $\langle 0, 1\rangle$—are Nash equilibria. In each monomorphic state, no single individual stands to gain by unilaterally changing her strategy. Yet, the monomorphic population state $\langle 1, 0\rangle$ composed of only a-types is not stable: it is a source, and the phase portrait reveals that the slightest perturbation will send the population straight off toward strategy b corresponding to the population state $\langle 0, 1\rangle$.

Sources represent an extreme case of instability. No source can be asymptotically stable, and so, by our criteria, we can conclude that no source can be evolutionarily significant. Yet, NE may well be sources. This fact suffices to demonstrate that the Nash equilibrium is too weak a concept. This line of thinking provides the motivation for the following refinements of the NE concept that strengthen its requirements with

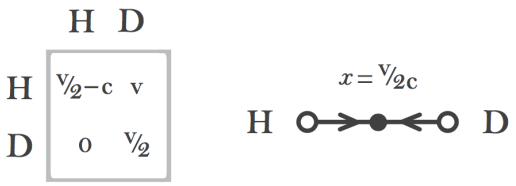


FIGURE 7. The mixed state state $\langle 1 - v/2c, v/2c \rangle$ cannot be a strict Nash equilibrium of the game, but is asymptotically stable.

the aim of narrowing in on the subset of viable equilibria by eliminating implausible fixed points.

We may strengthen the Nash equilibrium concept to the strict Nash equilibrium. The strict NE has the virtue of eliminating all the unstable states captured by the plain NE. Sources cannot be strict NE. However, the strict NE is clearly too strong, and eliminates crucial subsets of NE that can be evolutionarily significant, namely, the set of mixed strategy Nash equilibria.

To define the Nash equilibrium we simply modify definition 5, of the NE for 2×2 symmetric normal form games, by making the inequality between payoffs strict, as follows:

Definition 6 (Strict Nash Equilibrium)**.** *A state $x \in X$ is a strict Nash equilibrium if, for all $y \in \Delta$ where $y \neq x$, $u(x, x) > u(y, x)$.*

It is a basic result of game theory a mixed strategy cannot be a strict Nash equilibrium of a game (Osborne 2000). Yet mixed strategies can be asymptotically stable under the replicator dynamics. To see this, we need only consider any anti-coordination game, as in the Hawk-Dove game in FIGURE 7. As the phase portrait reveals, the polymorphic state represented by the mixed strategy equilibrium is indeed both stable and attracting. Any perturbation from the interior fixed point will result in a return to that point. This is an asymptotically stable mixed strategy NE. Anti-coordination games are an important class of interaction structures, which explain the evolution of heterogeneity and complementarity, as in the cases of ritualized animal conflict (Maynard Smith and Price 1973), sex ratios (Hamilton 1967), and fairness norms (Skyrms 1992b).

Given the existence of asymptotically stable mixed NE, and the prevalence of inter-specific variation in nature, the exclusion of mixed strategy equilibria provides sufficient justification for eschewing reliance on strict NE in our analysis of evolutionary games.

7.2. **The Limits of the Evolutionarily Stable Strategy.** Introduced by Maynard Smith and Price (1973), the evolutionarily stable strategy (ESS) is the flagship equilibrium concept of evolutionary games. When game theory texts introduce evolutionary game theory, they typically do so by presenting a primer on ESS (Osborne 2000). When ecologists employ game theoretic techniques to explain the persistence of a given trait

in a population, they often do so by identifying the ESS of the interaction (Apaloo et al. 2009). Yet, as we will see, the ESS has significant shortcomings. While a substantive improvement over the normal and strict NE, the ESS concept still fails to capture many evolutionarily significant outcomes.

In a symmetric normal form game, an ESS is a strategy with the following property: a population in which nearly all individuals play this strategy is resistant to invasions by small groups of mutants playing any alternative strategy. Formally,

Definition 7 (Evolutionarily Stable Strategy)**.** *A state x is evolutionarily stable if, for all $y \in \Delta$ where $y \neq x$, there exists an $\bar{\epsilon}(y) > 0$ such that $(1-\epsilon)\cdot(x,x) + \epsilon\cdot(x,y) > (1-\epsilon)\cdot(y,x) + \epsilon\cdot(y,y)$ for $0 < \epsilon < \bar{\epsilon}(y)$.*

In an ESS, the expected payoffs to the incumbent strategy x is strictly greater than that of any alternate strategy y, used by a sufficiently small proportion ϵ of mutants in the population. Equivalently, in a way that makes its relation to the NE clear, the ESS may be formulated in terms of the combination of two conditions:

Definition 8 (Evolutionarily Stable Strategy)**.** *A state x is evolutionarily stable if, for all $y \in \Delta$ where $y \neq x$,*

(1) *$u(x,x) \geq u(y,x)$, and*
(2) *if $u(x,x) = u(y,x)$, then $u(x,y) > u(y,y)$.*

The first condition states that the incumbent strategy x composes a (weak) NE with itself, and the second condition says that if there is a mutant y who fairs as well against x as it does against itself, then x must do better against that mutant that it does against itself. That is, any mutant strategy must not be able to invade, and moreover must also be driven to extinction.

When compared to the strict NE, the ESS represents a clear move in the right direction. As with strict NE, with ESS we retain the guarantee of asymptotic stability, but unlike the strict NE, the ESS doesn't automatically disqualify mixed strategy equilibria.

In one direction, we have that if a state is an ESS, then it is indeed asymptotically stable. In the other direction, however, the implication does not hold. There are attracting states, and attracting sets of states that are not ESS. There are stable states and stable sets of states that are not ESS. And there are AS states, and collectively AS sets of states that are not ESS (Bomze 1983, 1995).

We consider examples of non-ES states and sets of states, as these are both attracting and stable. The reason that a stable attractor may not be an ESS is that the dynamics may spiral in elliptically towards the attractor. This cannot be covered by the notion of ESS due to the linearity inherent in its definition. To see this, we consider the game in FIGURE 8, where we have a global attractor that is not an ESS. Here, the point $x = \langle 1/3, 1/3, 1/3\rangle$ is no ESS, but globally attracting. It is readily observed that any interior state will converge to x, and we can confirm that x is not an ESS by working

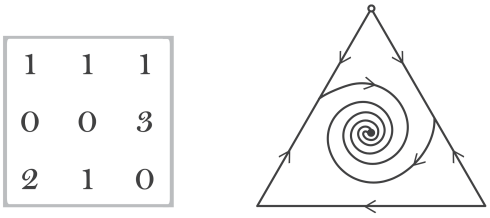


FIGURE 8. A globally attracting non-ESS.

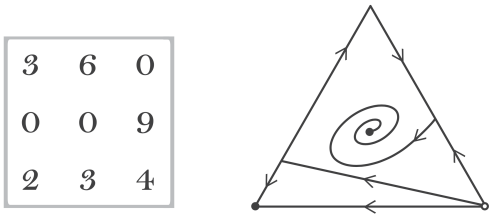


FIGURE 9. Zeeman's 1981 example of an attracting non-ESS.

through Definition 8 of ESS. First we observe that if one individual uses x then the other player gets an expected payoff of 1 no matter what strategy they use. That is, for any alternate strategy y, $u(x,x) = 1 = u(y,x)$. So condition (1) of ESS is satisfied, and no strategy does better against x than it does against itself. But this payoff tie means that we need to check the condition (2), which is indeed not satisfied.

To see this, consider a perturbed state such as $y = \langle 3/8, 1/4, 3/8\rangle$. Here, we get $u(x,y) \approx 1.031 > 0.96 \approx u(y,y)$ and so the mutant strategy y fares better against itself than x does against it. But the mutant's advantage is temporary as y is unstable and gives way to the invasion of another mutant strain, which gives way to another, and so on, with an orbit which spirals in towards x. A similar example of an attractor with a large basin of attraction is provided by Zeeman's 1981 counterexample depicted in FIGURE 9.

What these examples show is that, from the point of view of RD, an attractor is a more general notion than an ESS, and better characterization of resistance to mutation. Moreover, we have from (Zeeman 1981) that if an ESS lies in the interior of the state space, then its basin of attraction is the whole interior, and so it is the only attractor. But if an attractor is in the interior of the states space, it may have a smaller basin of attraction, and the game may still admit other attractors on the boundary of the state space (see FIGURE 9).

A defining limitation of the ESS concept is that it captures only single population states, and so cannot capture non-singleton sets of states, whatever their basins of attraction or stability properties; sets of states that are collectively asymptotically stable are beyond its reach. Consider the evolutionary game and dynamics in FIGURE 10.

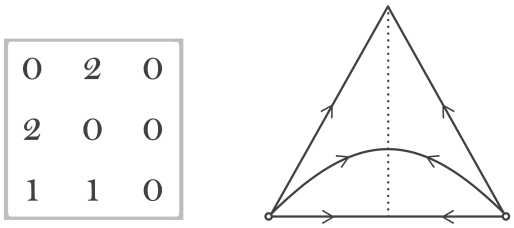


FIGURE 10. The set of states in $\{\langle a, b, c\rangle \,|\, a = b, c \in [0, 1]\}$ are collectively, but not individually, asymptotically stable.

Examining the phase portrait, it is clear that interior trajectories will converge to the set $E = \{\langle a, b, c\rangle \,|\, a = b, c \in [0, 1]\}$. But no member of E is an ESS. This is because every distinct strategy in this set is in a payoff tie against any other, and so the second condition of the ESS, that of strict inequality, will not be satisfied. Yet, when considered collectively, this set is both globally attracting and stable—we will see that it is an ESSet.

Why should we worry about sets of states qua sets? In normal form games, sets of fixed points tend not to be robust under small perturbations in the payoff structure of the game. Sets containing continua of equilibria can often be made to collapse to a single state. In extensive form games, however, this is not so. It is a well-known property of extensive form games with nontrivial structure that they produce non-singleton Nash sets (Cressman 2003). Importantly, it has been shown that in many extensive form games these sets are indeed robust under perturbations of payoffs (Huttegger 2010). Moreover, as in FIGURE 10, no single state in the attracting set is evolutionarily significant, since the set of initial conditions that will converge to any point in E is a measure zero set. It is only the set as a whole that commands convergence from a non-measure zero set of initial conditions, and so only the set as a whole qualifies as evolutionarily significant. Sets of states that are significant only as sets should be expected, particularly in extensive form games, and should be accounted for.

In short, the conditions provided by the ESS are clearly too strong in that they fail to capture (globally) attracting states, as well as some asymptotically stable states, and all collectively asymptotically stable sets of states.[13]

7.3. **The Limits of the Neutrally Stable Strategy.** The neutrally stable strategy (NSS) relaxes the ESS concept to weak inequality so as to include more evolutionarily significant outcomes. Particularly, it redeems some, though not all, of the Lyapunov stable states excluded by the ESS. An advantage of the NSS over the ESS is that it can account for members of significant Nash components, as well as portions of some limit cycles. In the latter case, the hubs of cycles can be NSS (e.g., in the RPS game), while

[13]For an analysis on asymptotically stable sets see (D'Aniello and Steele 2006).

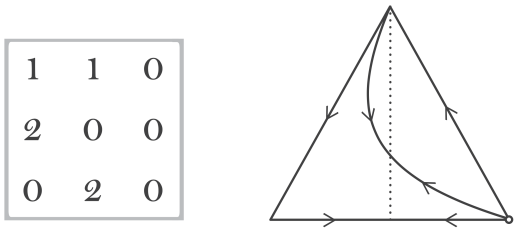


FIGURE 11. The set of states $\{\langle a, b, c\rangle \mid a = b > 0\}$ are Lyapunov stable, but not NSS.

the orbits themselves will go uncaptured. A basic property of an NSS under RD is that it is Lyapunov stable (Weibull 1997). The converse is not generally true, and we will see that there can be Lyapunov stable states that are not NSS. A NSS is not necessarily attracting, and so it is not necessarily asymptotically stable. There are evolutionarily insignificant NSS, as demonstrated by $n > 1$-player degenerate games $A = [0]$, where every point in the state space is an NSS with a measure-zero basin of attraction. The NSS captures some evolutionarily significant states neglected by the ESS at the expense of including other insignificant states.

Intuitively, at an NSS, mutant strategies will not be able to invade, but neither will they be driven out. Formally, the NSS is a weakening of the ESS to weak inequality in condition (2).

Definition 9 (Neutrally Stable Strategy)**.** *A state x is neutrally stable if, for all $y \in \Delta$ where $y \neq x$,*

(1) *$u(x, x) \geq u(y, x)$, and*
(2) *if $u(x, x) = u(y, x)$, then $u(x, y) \geq u(y, y)$.*

To see what significant states the NSS captures that the ESS does not, we revisit the game in FIGURE 10 with an asymptotically stable set, none of the members of which are ESS. Because of the relaxing of condition (2) to non-strict inequality in the NSS, all of the states that are eliminated by the ESS remain; the NSS captures each of the states belonging to the set. Again, however, NSS is defined for individual states and cannot account for the set qua set. This is unfortunate, since it is the set that is asymptotically stable, whereas any one of its members is only Lyapunov stable.

Let us consider an example demonstrating that the converse to the essential property of NSS does not hold—that not all Lyapunov stable states are NSS. Consider the game and state space in FIGURE 11. Here, all population states $x = \langle a, b, c\rangle$ with $a = b > 0$ are Lyapunov stable, but only those with $1/3 \leq a \leq 1/2$ are also NSS. Define the set of Laypunov stable states as $\Delta^{LS} = \{\langle a, b, c\rangle \mid a = b > 0, c \in [0, 1]\}$ and the set of NSS $\Delta^{NS} = \{\langle a, b, c\rangle \mid a = b > 0, a \in [1/3, 1/2], c \in [0, 1]\}$. Let x belong to Δ^{LS}, say $x = \langle 1/4, 1/4, 1/2\rangle$. Then $u(x, x) = 3/4$, and x is a NE with itself, however, it is not the case

that $u(x,y) \geq u(y,y)$ for all y. Consider $y = \langle 1,0,0\rangle$, then $u(y,y) = 1 > 3/4 = u(x,y)$. Thus x is Lyapunov stable but no NSS.[14]

7.4. **The Limits of the Evolutionarily Stable Set.** The Evolutionarily stable set (ESSets) was introduced by Bernard Thomas in (1985), and is the set-valued generalization of the ESS. In many senses, ESSets represents the best option on the table. The ESSet contains the ESS as a special case—the singleton ESSet—and so it captures all of the evolutionarily significant outcomes that the ESS does. On top of this, all non-singleton ESSets are also asymptotically stable, and so it captures evolutionarily significant outcomes excluded by the ESS. An ESSet can be a global attractor in a broader range of cases than the ESS. Specifically, if an ESSet set contains an interior NE, then it is globally asymptotically stable (Cressman 2003). Moreover, by construction, the ESSet fares no worse than the ESS in capturing evolutionarily insignificant outcomes.

Definition 10 (Evolutionarily Stable Set)**.** *A closed non-empty set of strategies E is an ESSet if and only if for each $x \in E$ there exists a neighborhood B around x such that, for all $y \in B$,*

(1) *If $y \in S$, then $u(x,y) \geq u(y,y)$, and*
(2) *If $y \notin S$, then $u(x,y) > u(y,y)$.*

All members of an ESSet are NSS (Weibull 1997), though there are NSS that will not belong to ESSets (Bomze 1983). From this it follows directly that there are Lyapunov stable states that are not in ESSets. But there are also non-NSS Lyapunov stable states that do not belong to an ESSet. Unfortunately, many of the asymptotically stable states that are not ESS are also not ESSets.

Now, we can account for the central outcome of the games like that in FIGURE 11. Recall that the set $E = \{\langle a,b,c\rangle \,|\, a = b, c \in [0,1]\}$ could not be captured by the ESS, and was only captured by the NSS in terms of its members, which are not themselves individually asymptotically stable. But E is an ESSet. To see this, let $x \in E$ and $y \notin E$. So x and y are of the form $\langle a,b,c\rangle$ and where $a = b$ and $a \neq b$ respectively. By looking at the game matrix, we see that by fixing the value of c, the game reduces to the upper left 2×2 subgame where the only symmetric Nash equilibrium is where $a = b$. Thus for any such $y \notin E$, there is an $x \in E$ such that $u(x,y) > u(y,y)$ as required by (1). On the other hand, if $y \in S$, by simple algebra it can be shown that $u(x,y) = u(y,y)$.

The games in FIGURES 8, 9—the globally attracting non-ES state, and Zeeman's 1981 counterexample—both also work as counterexamples for the ESSet. This follows from the fact that they are both singleton sets of equilibria which are not ESS. Nothing in the ESSet concept is added that can deal with them, and the same arguments as

[14]Such an example was first put forward by Bomze and Weibull (1995).

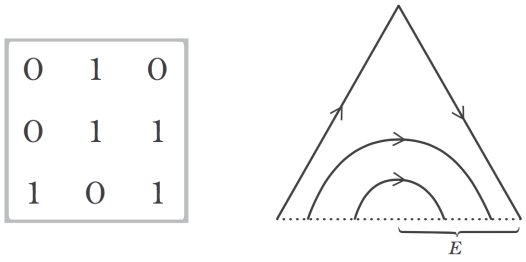


FIGURE 12. The set of states in $E = \{\langle a, b, 0\rangle \,|\, b > 1/2\}$ are all NSS, but do not belong to any ESSet.

those provided in the case of ESS carry through. Thus we still have asymptotically stable states that are not ESSets.

Moreover, we can also find non-singleton sets which are evolutionarily significant but not ESSets. In FIGURE 12 we see an of attracting set of NSS that does not belong to an ESSet. To see how this is so, consider the set $E = \{\langle a, b, 0\rangle \,|\, b \geq 1/2\}$. This is not an ESSet, because of the boundary point $x = \langle 1/2, 1/2, 0\rangle$. This can be observed in how the state $y = \langle 1/2 - \epsilon, 1/2 - \epsilon, 2\epsilon\rangle$ fares. We find that the payoff to x against y is $u(x, y) = -\epsilon$, while the payoff to y against itself is $u(y, y) = -\epsilon/2 - \epsilon^2$. For sufficiently small values of ϵ we get that $u(y, y) > u(x, y)$ violating condition (2), and making it so that E is no ESSet. Because a single boundary point of the set is not attracting, the set is as a whole technically not attracting. Yet we can anticipate that all interior trajectories will end up at the set of stable states comprising E. The set E then constitutes an evolutionarily significant outcome by our definition, but is not captured by the ESSet concept.[15]

8. DISCUSSION

In summary (SEE TABLE 1), we have seen that, under the replicator dynamics, the basic Nash equilibrium concept is far too weak. The NE captures all interior fixed points, but among these we find many unstable states (FIGURE 6). Its natural modification, the strict Nash equilibrium, is far too strong. The strict NE eliminates the unstable states captured by the NE, but fails to capture asymptotically stable polymorphic states (FIGURE 7). The flagship ESS concept is also too strong, and while it captures many of the AS polymorphic excluded by the strict NE, it fails to capture AS states where the dynamic flows elliptically towards the attractor (FIGURES 8,9), and collectively, but not individually, AS sets (FIGURE 10). The NSS is more inclusive than the ESS, but still fails to capture some stable states (FIGURE 11), as well as set-valued outcomes. The most sophisticated equilibrium concept, the ESSet, does capture many sets of

[15]Apaloo et al. (2009, 510) describe this in terms of the ESSet lacking a 'stable convergence' property.

Equilibrium concepts	TOO WEAK (includes non-ES outcomes)		TOO STRONG (excludes ES outcomes)		
	Unstable states	Non-attracting states	Sets of states (qua sets)	Asymptotically stable states	Limit cycles
Nash equilibrium	YES (FIG.6)	YES (FIG.6)	YES (FIG.10)	NO	YES (FIG.3)
Strict Nash equilibrium	NO	NO	YES (FIG.10)	YES (FIG.7,8,9)	YES (FIG.3)
Evolutionarily stable strategy	NO	NO	YES (FIG.10)	YES (FIG.8,9)	YES (FIG.3)
Neutrally stable strategy	NO	YES (FIG.6,11)	YES (FIG.10)	YES (FIG.8,9)	YES (FIG.3)
Evolutionarily stable set	NO	NO	NO	YES (FIG.8,9)	YES (FIG.3)

TABLE 1. Failings of the equilibrium concepts in capturing all and only *evolutionarily significant* (ES) outcomes of the replicator dynamics.

collectively asymptotically stable states, though it still fails to captures others (FIGURE 12), and falls prey to elliptically attracting AS states (FIGURES 8,9) just as the ESS does. When we assess the equilibrium concepts by their success in capturing all and only asymptotic stable sets of states, each comes up short.

Over and above these shortcomings, there is also a distinct in-principle argument for the claim that the equilibrium concepts are all too strong. The argument proceeds as follows: Equilibrium concepts are formulated exclusively in terms of states or sets of states. But there are outcomes that cannot be effectively expressed in these terms, for example limit cycles (FIGURE 3), and strange attractors (see [Skyrms (1992a,b)]). While not asymptotically stable, these are dependable regularities that can command convergence from a significant set of initial conditions of a system, hence we are compelled to say that they constitute evolutionarily significant outcomes. Thus, these constitute evolutionarily significant outcomes which are not predicted by the equilibrium concepts. Indeed, on this count equilibrium concepts are formulated in the wrong way to begin with. That is, this problem has no hope of being resolved by strengthening or weakening our concepts in just the right way. Any solution would necessarily involve reformulating the equilibrium concepts in very different terms. How to go about this appears to be a genuinely open question.

The challenge is that behavior, such as limit cycles and strange attractors are best-described in qualitatively different terms. Limit cycles are best described by procuring

the equations that detail their orbits (Cressman 2003), while describing the behavior of strange attractors, or even inferring their existence can demand the use of numerical methods (Skyrms 1992a,b). Given this, it may be unreasonable to expect the equilibrium or stability concepts to provide such descriptions in addition to their point- and set-valued predictions. It is reasonable, however, to take the presence of such evolutionarily significant disequilibrium behavior as one more reason to avoid exclusive reliance on equilibrium concepts, and to engage in practice of explicitly examining nontrivial dynamics on a game-by-game basis.

9. Conclusion

Our analysis of the results circumscribing the limitations of equilibrium concepts leaves us in an interesting place. We provided a novel formulation of evolutionary significance which makes explicits the interrelationship between evolutionary significance, stability concepts, and equilibrium concepts. This has the added virtue of enabling the comparison of predictions between different dynamics. In assessing equilibrium concepts, we have shown how stability concepts attempt to stand in for evolutionary significance, so that a determination of the agreement between the predictions of an equilibrium concept and a stability concept is a proxy for the determination of the agreement between the equilibrium concept and evolutionary significance. In addition, we have distinguished two distinct arguments against the exclusive use of equilibrium concepts. The first is consists in the demonstration that, even under favorable assumptions, each equilibrium concept is either too strong or too weak. The second is an in-principle argument that observes that there are evolutionarily significant outcomes, such as limit cycles, which cannot be effectively formulated in terms of states or sets of states, and so cannot be effectively expressed by equilibrium concepts.

We have not demonstrated that an equilibrium concept at the level of the payoff matrix must fail in capturing all and only evolutionarily significant outcomes, but we have provided strong arguments indicating that we should not expect a member of the family of Nash refinements can do the job. Our moral is that equilibrium concepts on their own are typically going to be unreliable tools for the analysis of dynamical processes, and that a more complicated, but also more interesting picture emerges from explicit investigation of the underlying dynamics.

References

Apaloo, J., J. S. Brown, and T. L. Vincent (2009). Evolutionary game theory: ESS, convergence stability, and NIS. *Evolutionary Ecology Research 11*(4), 489–515.

Bergstrom, C. T. and P. Godfrey-Smith (1998). On the Evolution of Behavioral Heterogeneity in Individuals and Populations. *Biology and Philosophy 13*, 205–231.

Bomze, I. M. (1983). Lotka-Volterra equation and replicator dynamics: A two-dimensional classification. *Biological Cybernetics 48*(3), 201–211.

Bomze, I. M. (1995, apr). Lotka-Volterra equation and replicator dynamics: new issues in classification. *Biological Cybernetics 72*(5), 447–453.

Bomze, I. M. and R. Burger (1995). Stability by mutation in evolutionary games. *Games and Economic Behavior 11*(2), 146–172.

Bomze, I. M. and J. W. Weibull (1995). Does neutral stability imply lyapunov stability? *Games and Economic Behavior 11*(2), 173–192.

Cressman, R. (2003). *Evolutionary Dynamics and Extensive Form Games*. MIT Press.

Cressman, R. and Y. Tao (2014). The replicator equation and other game dynamics. *Proceedings of the National Academy of Sciences 111*(Supplement_3), 10810–10817.

D'Aniello, E. and T. H. Steele (2006). Asymptotically stable sets and the stability of ω-limit sets. *Journal of Mathematical Analysis and Applications 321*(2), 867–879.

Fudenberg, D. and L. Imhof (2004). Stochastic Evolution as a Generalized Moran Process. *Unpublished ...*, 1–26.

Hamilton, W. D. (1967). Extraordinary Sex Ratios. *Science 156*(3774), 477–488.

Hofbauer, J. and S. M. Huttegger (2007). Selection-mutation dynamics of signaling games with two signals. *Language* (949), 0–3.

Hofbauer, J. and K. Sigmund (1998). *Evolutionary Games and Population Dynamics*.

Huttegger, S. (2010). Generic properties of evolutionary games and adaptationism. *The Journal of Philosophy 107*(2), 80–102.

Huttegger, S., B. Skyrms, P. Tarres, and E. Wagner (2014). Some dynamics of signaling games. *Proceedings of the National Academy of Sciences 111*(Supplement_3), 10873–10880.

Huttegger, S. M. and K. J. Zollman (2010). The Limits of ESS Methodology. In *Evolution and Rationality: Decisions, Co-operation and Strategic Behaviour*, pp. 67–83.

Huttegger, S. M. and K. J. Zollman (2013). Methodology in biological game theory. *British Journal for the Philosophy of Science 64*(3), 637–658.

Maynard Smith, J. and G. R. Price (1973). The logic of animal conflict. *Nature 246*(5427), 15–18.

Moran, P. A. (1962). *The Statistical Processes of Evolutionary Theory*. Clarendon Press, Oxford.

Osborne, M. J. (2000). *An Introduction to Game Theory*.

Sandholm, W. H. (2010). *Population Games and Evolutionary Dynamics*. The Mit Press.

Schuster, P. and K. Sigmund (1983). Replicator dynamics. *Journal of Theoretical Biology 100*(3), 533–538.

Skyrms, B. (1992a). Chaos and the Explanatory Significance of Equilibrium. *Proceedings of the Biennial Meeting of the Philosophy of Science Association*, 374–394.

Skyrms, B. (1992b). Chaos in game dynamics. *Journal of Logic, Language and Information 1*(2), 111–130.

Skyrms, B. (2000). Stability and Explanatory Significance of Some Simple Evolutionary Models. *Philosophy of Science 67*(1), 94–113.

Taylor, P. D. and L. B. Jonker (1978). Evolutionarily stable strategies and game dynamics. *Mathematical Biosciences 40*(1-2), 145–156.

Thomas, B. (1985). On evolutionarily stable sets. *Journal of Mathematical Biology 22*(1), 105–115.

Weibull, J. W. (1997). *Evolutionary Game Theory*. Evolutionary Game Theory. MIT Press.

Zeeman, E. C. (1981). Dynamics of the evolution of animal conflicts. *Journal of Theoretical Biology 89*(2), 249–270.

UC Irvine, Irvine, CA 92697, USA
E-mail address: amohseni@uci.edu